Module 1 Quiz - Solution Set

1. **What fundamental concepts of vector calculus are essential for machine learning optimization?**

   Answer: B) Gradients, partial derivatives, and directional derivatives

   Explanation: Gradients, partial derivatives, and directional derivatives are core to optimization in machine learning, helping understand how functions change and how to minimize loss functions.

2. **When optimizing machine learning models, what is the role of computing gradients?**

   Answer: B) To understand the curvature of the loss function

   Explanation: Gradients indicate how much the loss function will change with a small change in input values, allowing optimization algorithms to adjust model parameters for better performance.

3. **Which optimization algorithm is known for its iterative approach to finding the minimum of a function by moving in the direction of steepest descent?**

   Answer: B) Gradient descent

   Explanation: Gradient descent is the most common method for optimization, moving iteratively towards the minimum of a function by following the direction of the steepest slope.

4. **What is the primary motivation behind using stochastic gradient descent (SGD) instead of traditional gradient descent?**

   Answer: C) SGD is less computationally intensive

   Explanation: SGD reduces computational complexity by working with mini-batches rather than the full dataset, leading to faster convergence.

5. **In machine learning, what does it mean to have an unconstrained optimization problem?**

   Answer: B) The optimization problem does not involve any constraints on the variables

Explanation: In unconstrained optimization, the variables are free to take any values that minimize the objective function.

6. **Which optimization technique is used to find both local or global minima in unconstrained scenarios?**

   Answer: C) Simulated annealing

   Explanation: Simulated annealing is a probabilistic method used for finding global minima in complex spaces, making it suitable for unconstrained scenarios.

7. **What is the purpose of Lagrange multipliers in constrained optimization problems?**

   Answer: A) To enforce constraints while optimizing the objective function

   Explanation: Lagrange multipliers allow the inclusion of equality constraints in the optimization process.

8. **When dealing with constrained optimization in machine learning, what trade-offs must be considered?**

   Answer: C) Between computational efficiency and convergence rate

   Explanation: Optimizing while considering constraints requires a balance between finding an optimal solution and the time and resources needed to achieve convergence.

9. **Which optimization algorithm moves towards the minimum of a function in discrete steps, making it suitable for large datasets?**

   Answer: B) Mini-batch SGD

   Explanation: Mini-batch SGD updates the model parameters using small batches of data, combining the advantages of both gradient descent and stochastic gradient descent.

10. **Solve the following problem and confirm the point you are finding as candidate is**

$$\min \ f(x_1, x_2) = \frac{1}{2}(x_1^2 + 4x_2^2) - x_1 - 2x_2 \quad \text{subject to} \quad x_2 \geq 1$$

Solution Using Karush-Kuhn-Tucker (KKT) Conditions

Define the Lagrangian function:

$$\mathcal{L}(x_1, x_2, \mu) = \frac{1}{2}(x_1^2 + 4x_2^2) - x_1 - 2x_2 + \mu(1 - x_2)$$

where μ is the Lagrange multiplier for the inequality constraint.

The KKT conditions are:

(a) $\dfrac{\partial \mathcal{L}}{\partial x_1} = x_1 - 1 = 0$

(b) $\dfrac{\partial \mathcal{L}}{\partial x_2} = 4x_2 - 2 - \mu = 0$

(c) $x_2 \geq 1, \quad \mu \geq 0$ (primal and dual feasibility)

(d) $\mu(1 - x_2) = 0$ (complementary slackness)

We proceed by cases based on the complementary slackness condition.

Case 1: $\mu = 0$ (Constraint is inactive, $x_2 > 1$)

From condition 1: $x_1 - 1 = 0 \implies x_1 = 1$

From condition 2: $4x_2 - 2 - 0 = 0 \implies 4x_2 = 2 \implies x_2 = \frac{1}{2}$

However, $x_2 = \frac{1}{2}$ violates the primal feasibility condition $x_2 \geq 1$. Therefore, this case is invalid.

Case 2: $x_2 = 1$ (Constraint is active)

We have $x_2 = 1$.

From condition 1: $x_1 - 1 = 0 \implies x_1 = 1$

From condition 2: $4(1) - 2 - \mu = 0 \implies 4 - 2 = \mu \implies \mu = 2$

Check dual feasibility: $\mu = 2 \geq 0$. **All KKT conditions are satisfied**.

Candidate Point and Objective Value

The candidate point is $(x_1, x_2) = (1, 1)$.

The value of the objective function at this point is:

$$f(1,1) = \frac{1}{2}(1^2 + 4 \cdot 1^2) - 1 - 2 \cdot 1 = \frac{1}{2}(5) - 3 = 2.5 - 3 = -\frac{1}{2}$$

Verification via Hessian

The Hessian matrix of the objective function is constant:

$$H(f) = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

The eigenvalues are 1 and 4, both positive. This means the objective function is **strictly convex** everywhere.

For a strictly convex function with a linear constraint, any point satisfying the KKT conditions is the **unique global minimum**.

Conclusion

The unique global minimum for the constrained problem is:

$$\boxed{x_1 = 1, \quad x_2 = 1, \quad f(1,1) = -\frac{1}{2}}$$